HR Schema Insights

Simple Queries for Beginners

Tank54 Group | Oracle HR Schema

This guide shows what useful information you can get from the HR database using simple queries. Every query uses only 1 or 2 tables. We use INNER JOIN, LEFT JOIN, RIGHT JOIN, FULL OUTER JOIN, and WHERE clause. Each query answers one simple business question.

English level: A1 (easy words, short sentences)

The Tables We Use

In this guide we use only 4 tables from the HR database. Here is what each table has:

Table	What It Has	Key Columns
EMPLOYEES	All workers in the company	first_name, last_name, salary, department_id, manager_id, hire_date, job_id, commission_pct
DEPARTMENTS	Company departments (teams)	department_id, department_name, manager_id, location_id
JOBS	Job titles (roles) and pay range	job_id, job_title, min_salary, max_salary
LOCATIONS	Office cities and countries	location_id, city, country_id

How Tables Connect:
- EMPLOYEES and DEPARTMENTS connect by DEPARTMENT_ID
- EMPLOYEES and JOBS connect by JOB_ID
- DEPARTMENTS and LOCATIONS connect by LOCATION_ID
- EMPLOYEES and EMPLOYEES connect by MANAGER_ID (a manager is also an employee)

Part 1: INNER JOIN Insights

INNER JOIN shows only rows that match in both tables. Use it when you want only rows that have a match on both sides.

Insight 1: Employee Name and Department

Business Question:

What is the name of each employee and which department do they work in?

SQL Query:

```
SELECT e.first_name,
     e.last_name,
     d.department_name
FROM employees e
INNER JOIN departments d
     ON e.department_id = d.department_id
ORDER BY d.department_name;
```

How It Works:

We take EMPLOYEES and DEPARTMENTS. The shared column is DEPARTMENT_ID. INNER JOIN shows only employees who have a department. If an employee has no department, they do not appear. The result has two names from two tables, but in one row. This is the most common JOIN query in real work.

Result:

First Name	Last Name	Department
Diana	Lorentz	IT
Steven	King	Executive
Neena	Kochhar	Executive
Shelley	Higgins	Accounting

Remember: "e" is a short name for employees. "d" is a short name for departments. We use short names to make the query shorter and easier to write.

Insight 2: Employee Name and Job Title

Business Question:

What is the name and job title (role) of each employee?

SQL Query:

```
SELECT e.first_name,
     e.last_name,
     j.job_title
FROM employees e
INNER JOIN jobs j
     ON e.job_id = j.job_id
ORDER BY j.job_title;
```

How It Works:

We take EMPLOYEES and JOBS. The shared column is JOB_ID. Every employee has a job_id. So every employee appears in the result. This query shows what role each person has in the company. For example: Steven is the President, Diana is a Programmer.

Result:

First Name	Last Name	Job Title
Steven	King	President
Neena	Kochhar	Administration Vice President
Diana	Lorentz	Programmer
Jennifer	Whalen	Administration Assistant

Insight 3: Department Name and City

Business Question:

Where is each department located? Show the department name and the city.

SQL Query:

```
SELECT d.department_name,
     l.city
FROM departments d
INNER JOIN locations l
     ON d.location_id = l.location_id
ORDER BY l.city;
```

How It Works:

We take DEPARTMENTS and LOCATIONS. The shared column is LOCATION_ID. Each department has a location_id. So every department appears. This shows which city each team works in. For example: IT is in South San Francisco, Executive is in Seattle.

Result:

Department	City
IT	South San Francisco
Executive	Seattle
Sales	Seattle
Marketing	Toronto

Insight 4: Employees in One Department (WHERE)

Business Question:

Who works in the IT department? Show their name and salary.

SQL Query:

```
SELECT e.first_name,
     e.last_name,
     e.salary
FROM employees e
INNER JOIN departments d
     ON e.department_id = d.department_id
WHERE d.department_name = 'IT';
```

How It Works:

First we join EMPLOYEES with DEPARTMENTS to get the department name. Then we use WHERE to keep only the rows where department_name is "IT". WHERE is a filter. It keeps only the rows you want. You can change "IT" to any other department name to see its employees.

Result:

First Name	Last Name	Salary
Alexander	Hunold	9000
Bruce	Ernst	6000
David	Austin	4800
Valli	Pataballa	4800
Diana	Lorentz	4200

Tip: You can change 'IT' to any department name to see employees in that department. For example: 'Sales', 'Executive', 'Marketing'.

Insight 5: Employees With High Salary (WHERE)

Business Question:

Which employees earn more than 10000? Show their name, salary, and job title.

SQL Query:

```
SELECT e.first_name,
     e.last_name,
     e.salary,
     j.job_title
FROM employees e
INNER JOIN jobs j
     ON e.job_id = j.job_id
WHERE e.salary > 10000
ORDER BY e.salary DESC;
```

How It Works:

We join EMPLOYEES with JOBS to see the job title. WHERE e.salary > 10000 keeps only employees whose salary is bigger than 10000. ORDER BY e.salary DESC shows the highest salary at the top. DESC means "from big to small". You can change 10000 to any number to find different salary levels.

Result:

First Name	Last Name	Salary	Job Title
Steven	King	24000	President
Neena	Kochhar	17000	Administration VP
Lex	De Haan	17000	Administration VP
Shelley	Higgins	12008	Accounting Manager

Tip: You can use other operators in WHERE:
- e.salary < 5000 (less than 5000)
- e.salary >= 10000 (10000 or more)
- e.salary != 24000 (not equal to 24000)
- e.salary BETWEEN 5000 AND 10000 (between 5000 and 10000)

Part 2: LEFT JOIN Insights

LEFT JOIN shows ALL rows from the left table (the first table). If a row has no match in the right table, it shows null. Use it when you want to keep ALL rows from the first table.

Insight 6: All Employees With Their Department

Business Question:

Show ALL employees and their department. Include employees who have no department.

SQL Query:

```
SELECT e.first_name,
     e.last_name,
     d.department_name
FROM employees e
LEFT JOIN departments d
     ON e.department_id = d.department_id
ORDER BY d.department_name;
```

How It Works:

We use LEFT JOIN because we want ALL employees. EMPLOYEES is the left table (after FROM). If an employee has no department, we still see the employee name but the department name will be null. Compare this with Insight 1 (INNER JOIN). INNER JOIN hides employees with no department. LEFT JOIN keeps them.

Result:

First Name	Last Name	Department
Steven	King	Executive
Diana	Lorentz	IT
Kimberely	Grant	null

> Kimberely has no department. With INNER JOIN she is hidden. With LEFT JOIN she appears with null.

Insight 7: Employees Without a Department

Business Question:

Are there employees who have NO department? Show their name.

SQL Query:

```
SELECT e.first_name,
     e.last_name
FROM employees e
LEFT JOIN departments d
     ON e.department_id = d.department_id
WHERE d.department_id IS NULL;
```

How It Works:

First we use LEFT JOIN to keep all employees. If an employee has no department, the department_id from DEPARTMENTS will be null. Then WHERE d.department_id IS NULL keeps only those rows where the department is null. This means: "Show me only the employees who have no match in DEPARTMENTS." This is a data quality check. Every employee should have a department. If this query returns rows, there is a data problem to fix.

Result:

First Name	Last Name
Kimberely	Grant

> **Key Pattern:** LEFT JOIN + WHERE ... IS NULL = find rows with NO match.
> This is one of the most useful patterns in SQL.

Insight 8: Employees Hired After a Date (WHERE)

Business Question:

Show employees who started work after 1 January 2005. Show name, hire date, and department.

SQL Query:

```
SELECT e.first_name,
     e.last_name,
```

```
     e.hire_date,
     d.department_name
FROM employees e
LEFT JOIN departments d
     ON e.department_id = d.department_id
WHERE e.hire_date > '01-JAN-2005'
ORDER BY e.hire_date;
```

How It Works:

We use LEFT JOIN to keep all employees, even those without a department. WHERE e.hire_date > '01-JAN-2005' keeps only employees hired after that date. The date format in Oracle is DD-MON-YYYY. You can change the date to any date you want. This query helps HR see who is new in the company.

Result:

First Name	Last Name	Hire Date	Department
Kimberely	Grant	24-MAY-2007	null
Douglas	Grant	13-JAN-2008	Sales
Michael	Hartstein	17-FEB-2004	Marketing

Tip: You can also use these date comparisons:
- e.hire_date >= '01-JAN-2005' (on or after)
- e.hire_date < '01-JAN-2005' (before)
- e.hire_date BETWEEN '01-JAN-2005' AND '31-DEC-2006' (between two dates)

Part 3: RIGHT JOIN Insights

RIGHT JOIN shows ALL rows from the right table (the second table). If a row has no match in the left table, it shows null. Use it when you want to keep ALL rows from the second table.

Insight 9: All Departments and Their Employees

Business Question:

Show ALL departments and the employees who work there. Include departments with no employees.

SQL Query:

```
SELECT d.department_name,
     e.first_name,
     e.last_name
FROM employees e
RIGHT JOIN departments d
     ON e.department_id = d.department_id
ORDER BY d.department_name;
```

How It Works:

We use RIGHT JOIN because we want ALL departments. DEPARTMENTS is the right table (after JOIN). If a department has no employees, we still see the department name but the employee name will be null. This shows us which departments are empty (have no people). This is useful because an empty department costs money but has no workers.

Result:

Department	First Name	Last Name
Executive	Steven	King
Executive	Neena	Kochhar
IT	Alexander	Hunold
Payroll	null	null

> Payroll department has no employees. With RIGHT JOIN it appears with null employee names. With INNER JOIN it would be hidden.

Insight 10: Departments With No Employees

Business Question:

Which departments have ZERO employees?

SQL Query:

```
SELECT d.department_name
FROM employees e
RIGHT JOIN departments d
     ON e.department_id = d.department_id
WHERE e.employee_id IS NULL;
```

How It Works:

RIGHT JOIN keeps ALL departments. If a department has no employees, the employee_id will be null. WHERE e.employee_id IS NULL filters to show only departments where no employee was found. This tells management which departments exist but have no people working in them.

Result:

Department Name
Payroll
Treasury
Corporate Tax
Control And Credit
Shareholder Services
Benefits

> **Key Pattern:** RIGHT JOIN + WHERE ... IS NULL = find rows in the RIGHT table with NO match.

Insight 11: All Departments in One City

Business Question:

Show all departments in the city of Seattle.

SQL Query:

```
SELECT d.department_name
FROM employees e
RIGHT JOIN departments d
     ON e.department_id = d.department_id
WHERE d.location_id = 1700;
```

How It Works:

We use RIGHT JOIN to keep all departments. WHERE d.location_id = 1700 filters to keep only departments in Seattle. This shows us which departments have an office in Seattle. This is a simple filter on the DEPARTMENTS table.

Result:

Department Name
Executive
Sales
Finance
Accounting

Part 4: FULL OUTER JOIN Insights

FULL OUTER JOIN shows ALL rows from both tables. Nothing is hidden. Matched rows and unmatched rows all appear.

Insight 12: All Employees and All Departments

Business Question:

Show ALL employees and ALL departments. Match them together where possible.

SQL Query:

```
SELECT e.first_name,
     e.last_name,
     d.department_name
FROM employees e
FULL OUTER JOIN departments d
     ON e.department_id = d.department_id
ORDER BY d.department_name;
```

How It Works:

FULL OUTER JOIN shows everything. If an employee has no department, we see the employee with null department. If a department has no employees, we see the department with null employee. If both match, we see both names together. This is the most complete view. No row is hidden.

Result:

First Name	Last Name	Department
Steven	King	Executive
Kimberely	Grant	null
null	null	Payroll
null	null	Treasury

Insight 13: Data Problems: Missing Links

Business Question:

Show only the problem rows. Employees with no department AND departments with no employees.

SQL Query:

```
SELECT e.first_name,
     e.last_name,
     d.department_name
FROM employees e
FULL OUTER JOIN departments d
     ON e.department_id = d.department_id
WHERE e.employee_id IS NULL
   OR d.department_id IS NULL;
```

How It Works:

FULL OUTER JOIN shows all rows from both tables. Then WHERE filters to keep only the problem rows. e.employee_id IS NULL means the department has no employees. d.department_id IS NULL means the employee has no department. OR means: show rows where EITHER condition is true. This is the best data quality check because it finds problems on both sides in one query.

Result:

First Name	Last Name	Department
Kimberely	Grant	null
null	null	Payroll
null	null	Treasury

Key Pattern: FULL OUTER JOIN + WHERE IS NULL + OR = find ALL unmatched rows from BOTH tables.
This is the best way to find data problems.

Part 5: SELF JOIN Insights

A SELF JOIN is when a table joins with itself. In the EMPLOYEES table, the MANAGER_ID column points to another employee. So we can use SELF JOIN to find who is the manager of each person.

Insight 14: Employee and Their Manager

Business Question:

Show each employee and the name of their manager.

SQL Query:

```
SELECT e.first_name AS employee,
     m.first_name AS manager
FROM employees e
LEFT JOIN employees m
     ON e.manager_id = m.employee_id
ORDER BY e.last_name;
```

How It Works:

We use the EMPLOYEES table two times. "e" is for the employee. "m" is for the manager. ON e.manager_id = m.employee_id means: "find the manager by matching the employee manager_id with another employee employee_id". We use LEFT JOIN because the top boss (the President) has no manager. With LEFT JOIN, the President will appear with null for manager. AS gives a clear name to each column.

Result:

Employee	Manager
Steven	null
Neena	Steven
Lex	Neena
Alexander	Neena
Diana	Alexander

Steven is the President. He has no manager, so his manager is null. Every other employee has a manager.

Insight 15: Manager and Team Size

Business Question:

How many people report to each manager?

SQL Query:

```
SELECT m.first_name AS manager,
     COUNT(e.employee_id) AS team_size
FROM employees m
LEFT JOIN employees e
     ON m.employee_id = e.manager_id
GROUP BY m.first_name
ORDER BY team_size DESC;
```

How It Works:

"m" is the manager. "e" is the employee. ON m.employee_id = e.manager_id means: find all employees whose manager_id is this manager. COUNT counts how many employees report to each manager. GROUP BY m.first_name groups the count by manager name. ORDER BY team_size DESC shows the biggest team first. Managers with no team show 0 because of LEFT JOIN.

Result:

Manager	Team Size
Neena	5
Shelley	3
Steven	2
Alexander	4

Remember: GROUP BY + COUNT = count how many rows are in each group.
This pattern is very useful for reports.

Quick Summary

#	Insight	JOIN	WHERE	Tables
1	Employee + Department	INNER	No	EMP, DEPT
2	Employee + Job Title	INNER	No	EMP, JOB
3	Department + City	INNER	No	DEPT, LOC
4	One Department Employees	INNER	Yes	EMP, DEPT
5	High Salary Employees	INNER	Yes	EMP, JOB
6	All Employees + Dept	LEFT	No	EMP, DEPT
7	Employees Without Dept	LEFT	IS NULL	EMP, DEPT
8	New Hires After a Date	LEFT	Date	EMP, DEPT
9	All Depts + Employees	RIGHT	No	EMP, DEPT
10	Empty Departments	RIGHT	IS NULL	EMP, DEPT
11	Departments in One City	RIGHT	Yes	EMP, DEPT
12	All Emp + All Dept	FULL	No	EMP, DEPT
13	Data Problems	FULL	IS NULL + OR	EMP, DEPT
14	Employee + Manager	SELF LEFT	No	EMP, EMP
15	Manager Team Size	SELF LEFT	GROUP BY	EMP, EMP

Key Patterns to Remember

Pattern	What It Does
INNER JOIN	Shows only rows that match in both tables.
LEFT JOIN	Shows ALL rows from left table + matches from right.
RIGHT JOIN	Shows ALL rows from right table + matches from left.
FULL OUTER JOIN	Shows ALL rows from both tables.
SELF JOIN	Table joins with itself. Good for manager = employee.
WHERE column = value	Filter: keep only rows where column equals the value.
WHERE column IS NULL	Filter: keep only rows where column is empty (null).
WHERE column > value	Filter: keep only rows where column is bigger than value.
WHERE a IS NULL OR b IS NULL	Filter: keep rows where a is null OR b is null.
GROUP BY + COUNT	Count how many rows are in each group.

Most Useful Patterns for Beginners:

1. LEFT JOIN + WHERE IS NULL = find rows with no match (Insight 7)
2. RIGHT JOIN + WHERE IS NULL = find rows with no match in right table (Insight 10)
3. FULL OUTER JOIN + WHERE IS NULL + OR = find all data problems (Insight 13)
4. INNER JOIN + WHERE = join tables AND filter (Insight 4, 5)